

Aegis Series-7 Hydraulic Control System

Maintenance Manual

Document AEG-MM-700 | Revision 3

1. Scope

This manual covers scheduled maintenance, alarm response, and controller behavior for the Aegis Series-7 Hydraulic Control System (HCS). It supplements, and does not replace, the Operator Manual (AEG-OM-700).

3. Controller Reset Conditions

Do NOT reset the PLC-03 controller while hydraulic pressure is above 50 bar. The controller reset routine re-initializes valve driver outputs; if the circuit is still pressurized, this can produce a momentary uncommanded actuation of Isolation Valve IV-21.

Controller reset is only permitted when pressure, as reported by the active pressure sensor (PS-04 on revisions prior to 3.2, PS-04A on revision 3.2 and later), reads below 50 bar.

4. Alarm A17 Response

Alarm A17 (Hydraulic Pressure Low) may clear on its own within a few seconds if it was triggered by a transient pressure dip during valve transition. **If alarm A17 persists for more than 10 seconds, execute Shutdown Procedure 4.7** before investigating further. Do not attempt to clear a persistent A17 by silencing the alarm alone.

Shutdown Procedure 4.7: close IV-21, confirm pressure decay, set POWER selector to OFF, tag out per site lockout/tagout procedure, then proceed to Section 6 (Troubleshooting).

6. Troubleshooting: Low Discharge Pressure

If the HPU fails to reach normal operating pressure, check the following in order:

- Isolation valve IV-21 is closed or partially closed.
- Hydraulic fluid level is low.
- Pressure sensor (PS-04 / PS-04A) signal is invalid or out of calibration — see Alarm A19 in the Alarm Reference.

7. Component Access

The Hydraulic Power Pack (HPU) maintenance access panel is secured by four quarter-turn fasteners. Remove the panel only after the unit has been shut down per Section 4 and pressure has been bled per Section 8. Do not operate the unit with the panel removed.

9. Related Documents

Engineering Change Notice ECN-1042 (pressure sensor replacement and setpoint change), ECN-1058 (alarm threshold correction), and the Component Register (AEG-CR-700) should be consulted alongside this manual, as component identifiers and setpoints have changed across software revisions.